Module 1: Data Types, Variables, Operators and Type Casting

Set A

1. Café Billing

A newly opened café wants a simple billing application for its cashier. The cashier enters the price of three food items ordered by a customer. The system should calculate the total bill and display the amount after adding a GST of 5%.

Requirements:

● Read the prices of three food items.

● Calculate the subtotal.

● Calculate GST (5%).

● Display subtotal, GST, and final amount.

2. Student Attendance Percentage

A college wants to determine whether a student satisfies the minimum attendance requirement. The program should accept: total number of working days, and number of days attended. Calculate the attendance percentage and display it.

Note: Total days and days attended are whole numbers. Dividing two int values in Java performs integer division and truncates the decimal part. The program must explicitly typecast at least one operand to double before performing the division, so the percentage is calculated correctly.

Requirements:

● Accept integer inputs.

● Explicitly typecast an operand to double before division (do not rely on automatic promotion alone — write the cast).

● Compute the attendance percentage.

● Display the result with two decimal places.

3. Fuel Expense Estimator

A traveler plans to drive to another city. The travel application should estimate the fuel expense. The user enters: distance to be travelled (km), vehicle mileage (km/litre), and cost of one litre of petrol.

Requirements:

● Calculate fuel required (distance ÷ mileage).

● Calculate total fuel cost.

● Display both values clearly, with two decimal places.

Set B

1. Employee Salary Calculation

A startup company wants to automate salary calculation. The HR executive enters: Employee ID, Employee Name, Basic Salary. The company provides: HRA = 20% of Basic, DA = 15% of Basic, Professional Tax = ₹250.

Requirements:

● Use appropriate data types for ID, name, and salary values.

● Calculate Gross Salary (Basic + HRA + DA).

● Calculate Total Deduction and Net Salary.

● Display all computed values clearly.

2. Foreign Exchange Kiosk (Revised)

A foreign exchange kiosk converts Indian Rupees (INR) into US Dollars (USD) for travelers. The kiosk charges a service commission of 2% on the converted dollar amount, plus a flat processing fee of ₹50 (deducted from the INR amount before conversion).

The cashier enters: amount in INR to be converted, and the current exchange rate (INR per 1 USD).

Calculate and display: amount available for conversion after deducting the processing fee, converted amount in USD, commission charged (in USD), and the final USD amount handed to the customer.

Requirements:

● Use double for all currency and rate values.

● Explicitly typecast where int and double values are mixed, to avoid precision loss.

● Display all intermediate and final values rounded to two decimal places.

3. Rectangular Tank Capacity

A civil engineering company wants to estimate the amount of water a rectangular tank can hold. The user enters: length, breadth, and height.

Calculate: volume of the tank, and capacity in litres (1 cubic metre = 1000 litres).

Requirements:

● Use suitable data types (double) for dimensions.

● Perform explicit type conversion wherever necessary while converting volume to litres.

● Display both the volume and the capacity clearly.

Set C

1. Hospital Patient Billing

A hospital generates the patient's bill based on the following information — Input: Consultation Fee, Laboratory Charges, Pharmacy Charges, Room Charges. Constraints: Service Tax = 8%, Insurance covers 25% of the bill.

Display the final amount payable by the patient. The application should clearly display each calculation.

Requirements:

● Compute the subtotal from all four charge components.

● Apply Service Tax (8%) on the subtotal.

● Deduct Insurance coverage (25%) from the tax-inclusive bill.

● Display an itemized breakdown and the final payable amount.

2. Fixed Deposit Maturity Calculator (Revised)

A bank wants to calculate the maturity amount of a fixed deposit using the compound interest formula:

$A = P \times (1 + r/100)^n$

where P = Principal amount, r = annual interest rate (%), n = duration in years.

The customer enters: Principal amount (whole number of rupees), Annual interest rate (a decimal value, e.g., 6.5), and Duration of the deposit in years (a whole number).

Calculate and display: the Maturity Amount (A), and the Total Interest Earned (A − P).

Note: 'n' is entered as an int, but Math.pow() requires double arguments — the program must explicitly typecast n (and P, where required) to double before using them in Math.pow(). The interest rate divided by 100 must also be computed using double arithmetic, not integer arithmetic.

Requirements:

● Use Math.pow() for the exponent calculation.

● Explicitly typecast int values to double wherever they are mixed with decimal arithmetic.

● Display the maturity amount and interest earned rounded to two decimal places.

3. Restaurant Invoice with Coupon

A restaurant records the following details for every customer: Customer Name, Number of Adult Meals, Number of Children's Meals. Charges: Adult Meal = ₹280, Children's Meal = ₹160, Service Charge = 8%, GST = 5% (after service charge).

The restaurant also offers a coupon worth ₹250 if the total bill exceeds ₹3000.

Generate the final customer invoice showing: Food Cost, Service Charge, GST, Coupon Discount (if applicable), Final Amount Payable.

Note: This question includes one simple threshold check (bill > ₹3000) as an intentional light stretch beyond pure arithmetic, appropriate for the hard tier. If your teaching schedule has not yet reached Module 2 (Control Statements) at this point, this question may be deferred or replaced with a non-conditional variant.

Requirements:

● Compute Food Cost from the meal counts and rates.

- Apply Service Charge, then GST on the resulting amount.
- Apply the coupon discount conditionally, only if the bill exceeds ₹3000.
- Display a complete, itemized invoice